F1

$$y_f = y_0 + v_{0y}\Delta t - \frac{1}{2} g\Delta t^2$$

$$y_2 = H - \frac{1}{2} g(t_2)^2$$

$$y_3 = H - \frac{1}{2} g(t_3)^2$$

$$y_3 = H - \frac{1}{2} g(t_2 + 1)^2$$

$$y_2 - y_3 = 40$$

$$H - \frac{1}{2} g t_2^2 - \left(H - \frac{1}{2} g(t_2 + 1)^2\right) = 40$$

$$-\frac{1}{2} g t_2^2 + \frac{1}{2} g(t_2^2 + 2t_2 + 1) = 40$$

$$-\frac{1}{2} g t_2^2 + \frac{1}{2} g t_2^2 + g t_2 + \frac{g}{2} = 40$$

$$g t_2 + \frac{g}{2} = 40$$

$$t_2 = 3.58\ (seg)$$

$$y_3 = H - \frac{1}{2} g(t_2 + 1)^2$$

$$y_3 = H - \frac{1}{2} g(4.58)^2$$

$$y_4 = H - \frac{1}{2} g(t_2 + 2)^2$$

$$y_4 = H - \frac{1}{2} g(5.58)^2$$

$$\Delta y = y_3 - y_4$$

$$\Delta y = H - \frac{1}{2} g(4.58)^2 - \left[H - \frac{1}{2} g(5.58)^2\right]$$

$$\Delta y = -\frac{1}{2} g(4.58)^2 + \frac{1}{2} g(5.58)^2$$

$$\Delta y = 49.78\ m$$

F2

$$\vec{E} = \vec{A} + \vec{B}$$

$$\vec{E} = (5\hat{\imath} + 3\hat{\jmath}) + (-2\hat{\imath} + 4\hat{\jmath})$$

$$\vec{E} = 3\hat{\imath} + 7\hat{\jmath}$$

$$\vec{F} = \vec{C} - \vec{D}$$

$$\vec{F} = (-2\hat{\imath} - 3\hat{\jmath}) - (8\hat{\imath} - 2\hat{\jmath})$$

$$\vec{F} = -10\hat{\imath} - 1\hat{\jmath}$$

$$\vec{G} = \vec{E} \times \vec{F}$$

$$\vec{G} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 3 & 7 & 0 \\ -10 & -1 & 0 \end{vmatrix}$$

$$\vec{G} = [7 * 0 - (-1)(0)]\hat{\imath} - [3 * 0 - (-10)(0)]\hat{\jmath} + [3 * (-1) - (-10)(7)]\hat{k}$$

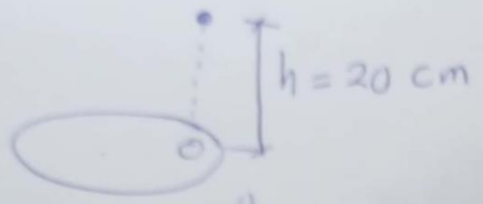
$$\vec{G} = 0\hat{\imath} - 0\hat{\jmath} + [-3 + 70]\hat{k}$$

$$\vec{G} = 0\hat{\imath} - 0\hat{\jmath} + 67\hat{k}$$

$$|G| = \sqrt{0^2 + 0^2 + 67^2}$$

$$|G| = 67$$

F3



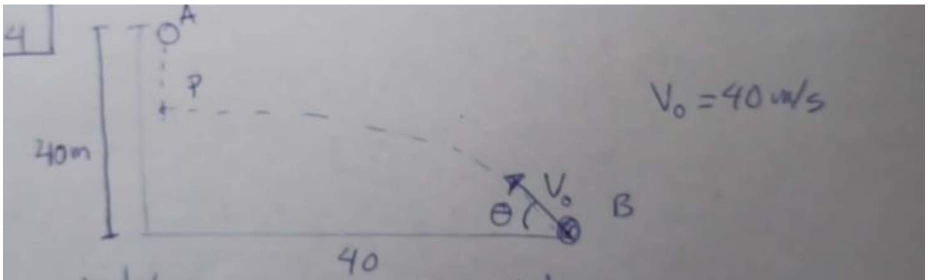

$$y = y_0 + \cancel{v_{0y}}^{\,0} t - \frac{1}{2} g t^2$$

$$0 = 0.2 - 5t^2$$

$$\underline{t = 0.2\ (s)}$$

$$W = \frac{2\pi}{0.2} = \underline{10\pi\ \text{rad/s}}$$

F4




Caida libre

$y = y_0 + \cancel{V_{0y}t}^{0} - \frac{1}{2}gt^2$

$y_A = 40 - 5t^2$

Parabolico

$y = \cancel{y_0}^{0} + V_{0y}t - \frac{1}{2}gt^2$

$y_B = 40\,\text{sen}\,\theta\, t - 5t^2$

$y_A = y_B$

$40 - \cancel{5t^2} = 40\,\text{sen}\,\theta\, t - \cancel{5t^2}$

$\cancel{40} = \cancel{40}\cos\theta\, t \quad \Big\} \; t = \frac{1}{\cos\theta}$

$\cancel{40} = \cancel{40}\,\text{sen}\,\theta\left(\frac{1}{\cos\theta}\right)$

$1 = \tan\theta \quad \therefore \; \theta = \tan^{-1}(1) \quad \theta = 45°$

F5

$\vec{V} = 48\hat{i} + 20\hat{j}$ $\quad h_{max} = 80$ m $\quad t_1 = ?$

para hallar $h_{max}$

$\cancel{V_{fy}^2}^{0} = V_{0y}^2 - 2g(Y_f - Y_i)$

$0 = V_{0y}^2 - 20(80)$

$V_{0y} = 40$ m/s

$V_{0x} = 48$ m/s

para hallar $t_1$

$V_{fy} = V_{0y} - g t_1$

$20 = 40 - 10t_1$

$\boxed{2 = t_1}$